

Subject Description Form

Subject Code	LSGI2375
Subject Title	Field Scheme (Surveying)
Credit Value	1
Level	2
Pre-requisite	LSGI2374 Surveying
Objectives	The aims of this project are: • To provide an opportunity for students to consolidate, integrate and practice the knowledge and skills learnt in subject areas of fundamental surveying, mapping and survey data analysis. • To enhance students' ability and skills in project planning and organization, logistics design and implementation, problem identification and solving, as well as working as a team. • To train the professionalism of the students in their work, especially in instrument calibration, carrying out field survey, taking field notes, checking and analysis of data, presentation of results, and caring of survey instruments. This subject emphasizes on team work, project management and problem solving in the field, hence will help students develop communication, critical and creative thinking, and cooperative attitudes and behavior of working with others.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Apply and integrate the knowledge learnt in Surveying b. Manage and carry out a survey project according to specifications and industrial best practices
Subject Synopsis/ Indicative Syllabus	The project work will involve control survey, topographical and detail survey, production of manual plans of survey and report writing.
Teaching/Learning Methodology	The subject has three phases. Firstly, students spend two days on campus developing survey standards that will meet the client's specification, checking equipment and arranging work schedules. Secondly, students will spend a week at the project site where students and supervisors are in residence together during which time all field data is collected. The task is to prepare a large-scale topographic map of an area with moderate variations in terrain elevation and having broad range of both natural and artificial features. The final phase is spent on campus where students finalise their reports and topographic map. Students usually work in groups. Each group will elect a group representative. The group representatives will meet with supervising staff every night to report on the group's progress and to discuss resource allocation and difficulties encountered. Supervisors can also use this time to give advice. Collection of field data is followed by the finalization of the topographic map, reports etc. Submission of which represents the completion of the survey camp.

	Students will be given feedbacks on-site of their performance and suggested improvements. Proficiency in equipment checking, operation, managing the project with limited supply of survey equipment, to carry out field observations, booking and checking, as well as data reduction and survey plan production will be assessed. This learning process will help students to develop communication skills, creative and critical thinking; to enable students to understand and appreciate the importance of team work; to provide students opportunities to develop resource, time and manpower management; and leadership skill.		
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/task	% weighting	Intended subject learning outcomes to be assessed
			a b
	1. Practicals	100	✓ ✓
	Total	100%	
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Student performance will be assessed based on their preparations, their field performance and the final product and report submitted at the end of the survey camp. Assessment areas will be based on the following: 1. Activities and documentation conforming to professional standards throughout; 2. Planning activity by each student, prior the period of field data collection, is acceptable to the staff supervisor of his/her group; 3. Processing of work to the satisfaction of the group supervisors; 4. Contribution to fieldwork, data reduction, plan production and report preparation by each student; 5. Fair drawn plots of the control scheme and survey, the referenced field data and computations, and a short report describing the methods used to complete the survey and produced by each student group at the end of period of residence at or near the project area.		
Student Study Effort Expected	Class contact:		
	▪ Practical work		40 Hrs.
	Other student study effort:		
	▪ Reading, data reduction and analysis, drafting of reports and topographic map.		5 Hrs.
	Total student study effort:		45 Hrs.

Reading List and References	Textbook: Schofield, W. (Wilfred), & Breach, M. (Mark). (2011). <i>Engineering surveying</i> (Sixth edition.). Spon Press. Bannister, A., Raymond, S. and Baker, R. (1998). <i>Surveying</i>. 7th Edition, Pearson Prentice Hall. Vosselman, G., Maas. H.-G. (2010). <i>Airborne and Terrestrial Laser Scanning</i>. Whittles Publishing. Armenakis, C., Patias, P. (2019). <i>Unmanned Vehicle Systems for Geomatics: Towards Robotic Mapping</i>. Whittles Publishing. Xing, M., Lu, Z., Yu, H. (2020). <i>InSAR and Data Processing</i>. MDPI – Multidisciplinary Digital Publishing. Dong, P., Chen, Q. (2017). <i>LiDAR Remote Sensing and Applications</i>. Taylor & Francis Group.
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